Predictive and Generative Modeling of Therapeutic Antibody Developability Using Fine-Tuned Protein Language Models

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BMI-702 Midterm Presentation

Background and Motivation

- Antibodies are Y-shaped proteins built from two chains:
  - **VH (Heavy-chain region)**: Contributes to antigen binding
  - **VL (Light-chain region)**: Pairs with VH, forms complete binding site

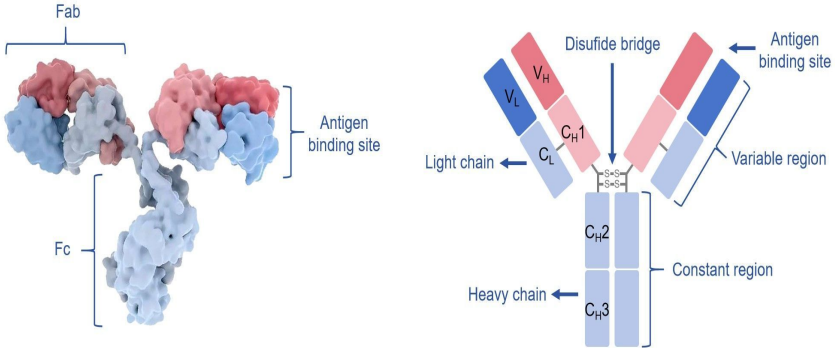
- Therapeutic antibodies are widely used but expensive and slow to develop

  ↳ Many candidates fail due to poor developability

  - Developability metrics include biophysical properties:
    - Thermostability (T_m)
    - Self-association propensity (AC-SINS)
    - Hydrophobicity (HIC, SAP)
    - Solubility

- Lead optimization focuses on modifying antibody sequences to improve these properties

  ↳ Requires balancing developability while preserving antibody binding function



Source: Creative Biolabs Antibody Development

Research Question: From Sequence to Developability

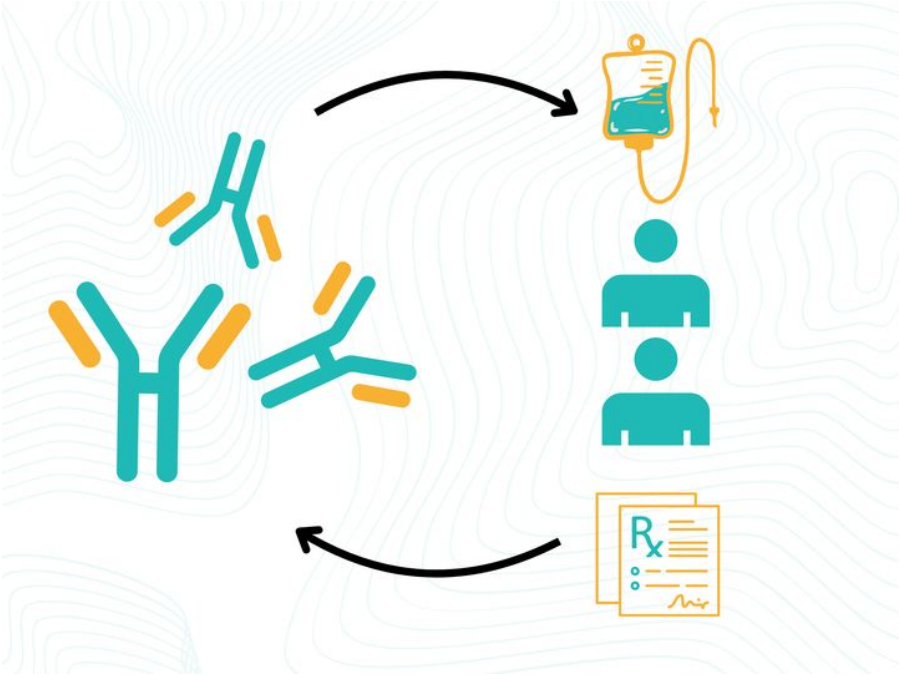
Main Research Question:

Can sequence-based protein language models learn representations that:

1. Predict developability properties and therapeutic approval?
2. Enable optimization of true developability properties through guided sequence generation?

- Prior work used experimentally measured features from antibody sequences to predict developability and downstream approval classification (Arsiwala et al., 2025)

- However, experimental features are not available early
  ↳Motivates the need for sequence-only ML models to predict developability



Source: Georgia Channing

Project Overview: Predict and Optimize Antibodies

Task A: Antibody Sequence Scoring

Input: Antibody sequence, VL and VH amino acid sequences

Output: Developability metrics and binary approval likelihood

Baseline Model: Ridge regression using p-IgGen for developability metrics, XGBoost for approval likelihood

Model: AbLang2 encoder -> embedding, with a classifier head (fine-tuned with LoRA)

Training Data: Ginkgo Bioworks PROPHET-Ab (GDPa1_v1.2) — 197 therapeutic antibodies

Task B: Antibody Sequence Generation

Input: Lead antibody CDR3, framework embedding, and desired HIC and AC-SINS target scores

Output: Generated CDR3 variants of the same length as the parent, with improved predicted developability (lower HIC, lower AC-SINS)

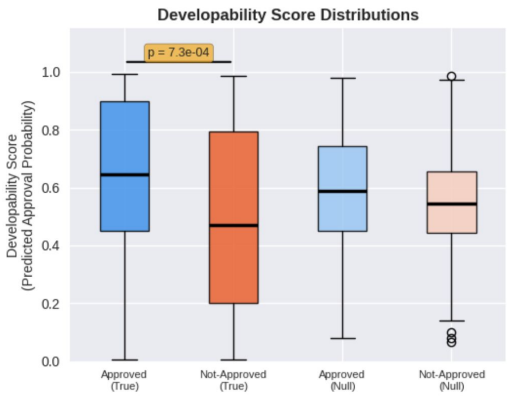
Model: Conditional flow matching in AbLang2 CDR3 PLS-reduced embedding space (480d → 10d), with FiLM conditioning on masked framework embeddings; AbLang2 MLM used as token-level sequence decoder, steered by flow model targets

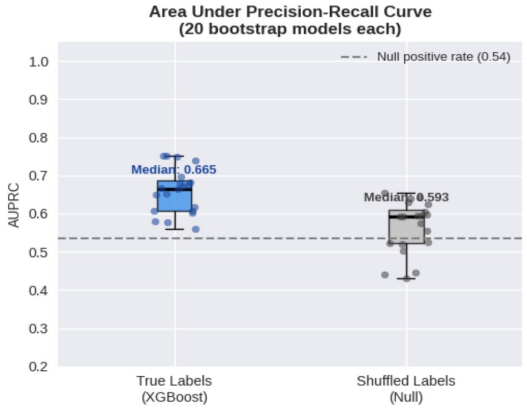
Training Data: Ginkgo Bioworks PROPHET-Ab (GDPa1_v1.2) — 197 therapeutic antibodies (folds 2–5)

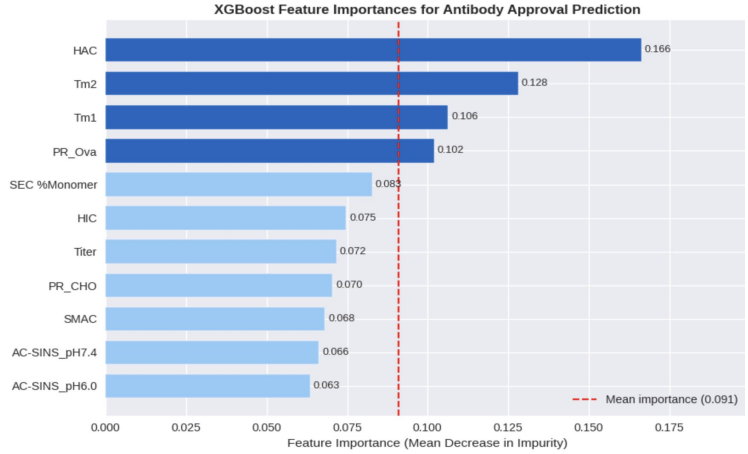
Therapeutic Approval Prediction Baseline Modelling

- **Data**: GDPa1 dataset with 197 antibodies, 30 raw feature columns
  - **Feature Selection**: 11 developability features (HIC, SMAC, HAC, PR_Ova, PR_CHO, SEC %Monomer, AC-SINS_pH6.0, AC-SINS_pH7.4, Tm1, Tm2, Titer)
  - **Binary classification**: 'Approved' (106 antibodies) vs. 'Not-Approved' (92 antibodies)
  - **Developability Score:** Per-sample mean predicted probability from XGBoost

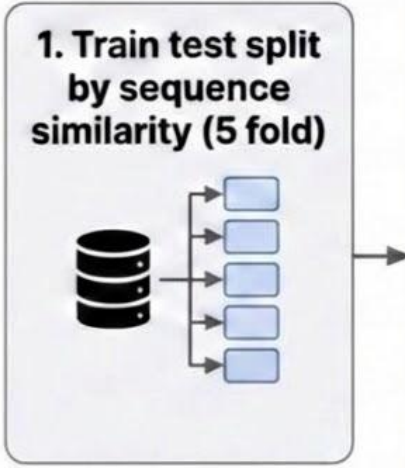
- Bootstrapped twenty independent XGBoost models trained on random 80:20 stratified train-test splits
  - Uses 11 experimentally measured biophysical features
  - Feature importances calculated by training a single model on all data

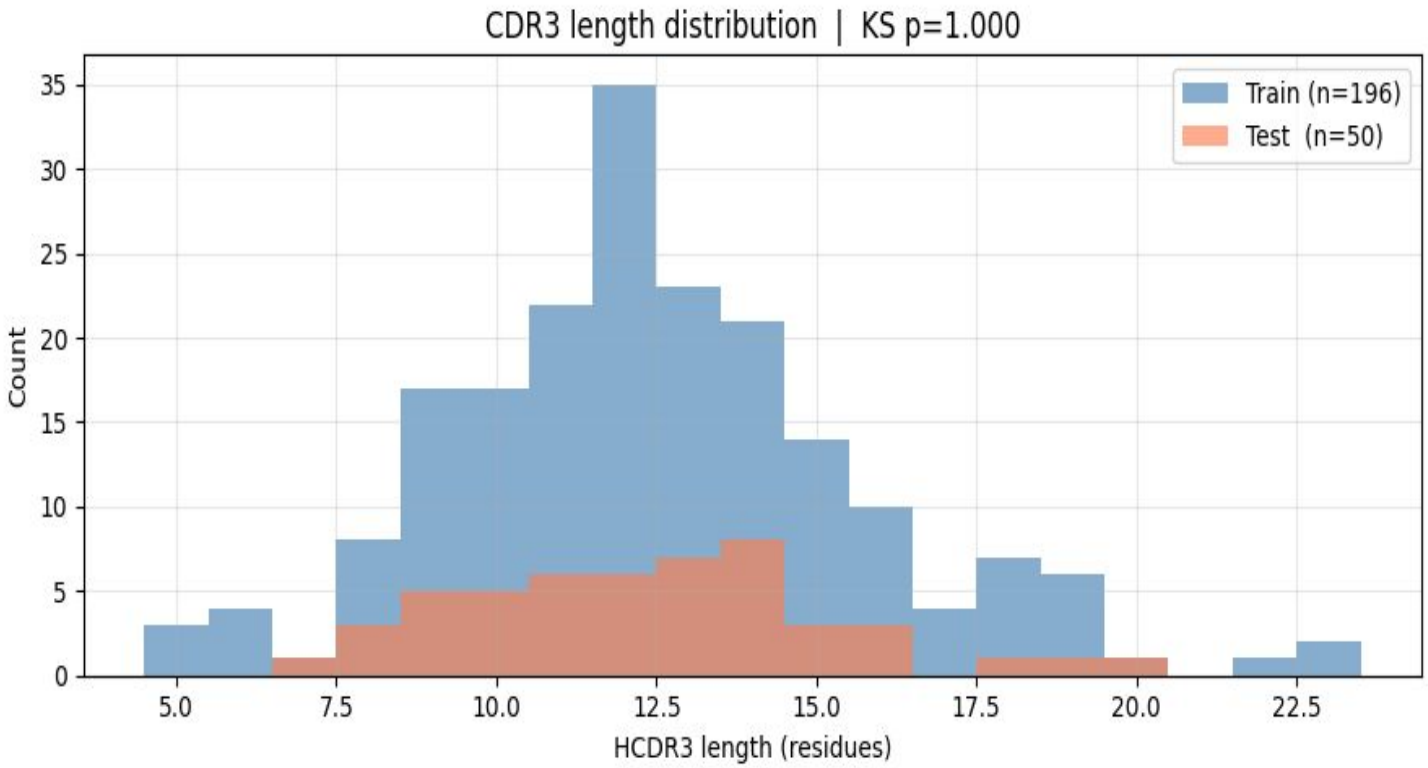






Generative Model Training Pipeline

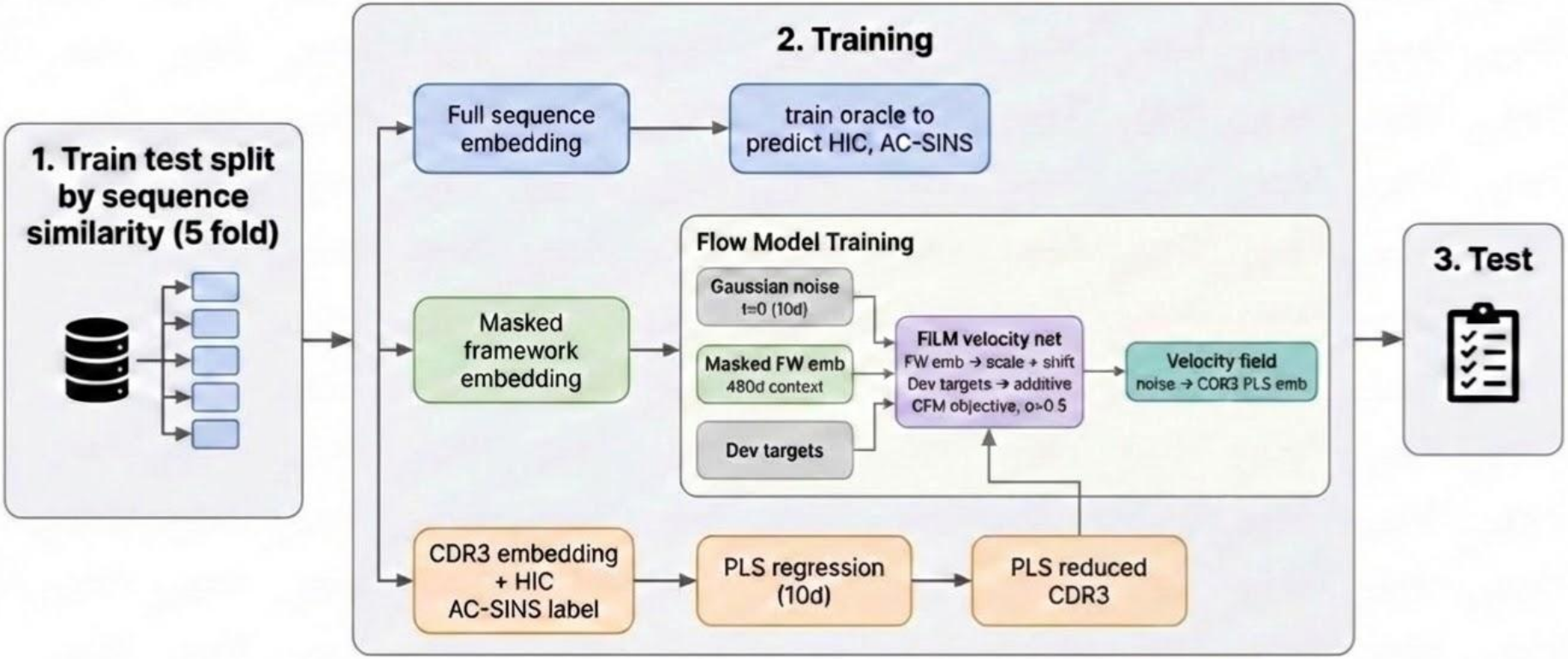




PLS: maximises covariance with HIC and AC-SINS (every dimension is developability relevant)

AC-SINS: Self-association
HIC: Hydrophobicity

Generative Model Training Pipeline



PLS: maximises covariance with HIC and AC-SINS (every dimension is developability relevant)

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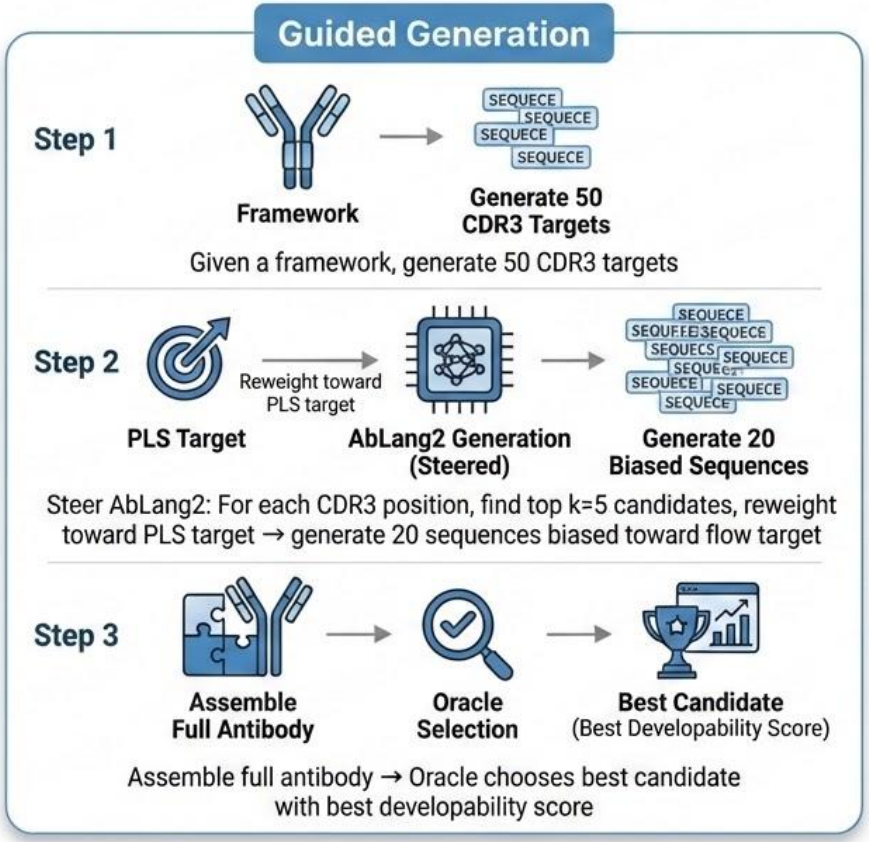
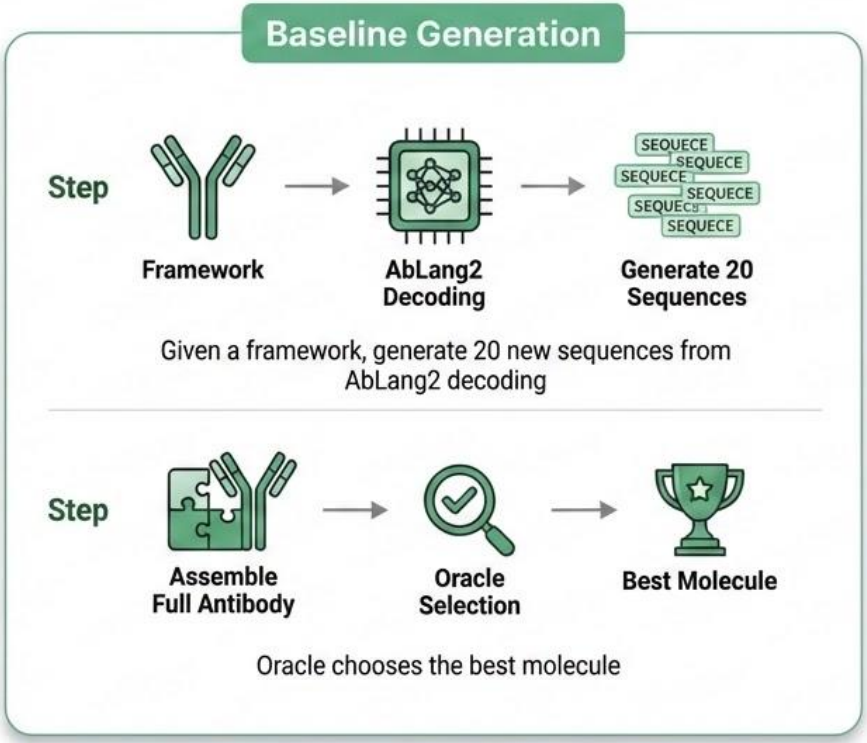
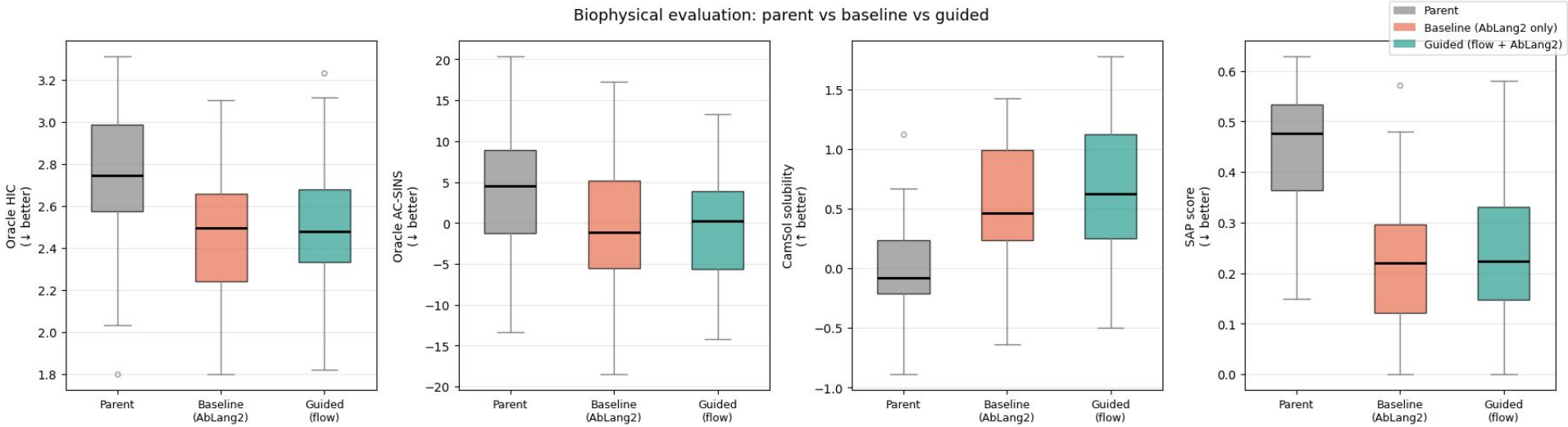
Generation - Guided with flow and Baseline (only AbLang2)





Result: AbLang2 Performs Well & Guided Generation generates favorable Abs



Metric	Parent	Baseline	Guided
Oracle HIC (↓)	+2.738±0.048	+2.466±0.043	+2.493±0.046
Oracle SINS (↓)	+3.741±1.215	−0.545±1.178	−0.441±1.044
CamSol (↑)	−0.004±0.056	+0.564±0.071	+0.649±0.078
SAP (↓)	+0.449±0.016	+0.216±0.019	+0.242±0.020

Paired t-tests (* = p < 0.05):

Metric	Baseline vs Parent	Guided vs Parent	Guided vs Baseline
Oracle HIC (↓)	Δ=−0.271 p=0.000*	Δ=−0.245 p=0.000*	Δ=+0.027 p=0.418
Oracle SINS (↓)	Δ=−4.286 p=0.000*	Δ=−4.182 p=0.000*	Δ=+0.104 p=0.872
CamSol (↑)	Δ=+0.568 p=0.000*	Δ=+0.653 p=0.000*	Δ=+0.085 p=0.463
SAP (↓)	Δ=−0.233 p=0.000*	Δ=−0.207 p=0.000*	Δ=+0.026 p=0.414

For ↓ metrics: Δ < 0 = improvement.
For ↑ metrics: Δ > 0 = improvement.

AC-SINS: Self-association
HIC: Hydrophobicity
CamSol: Computed solubility
SAP: Aggregation due to hydrophobic patches

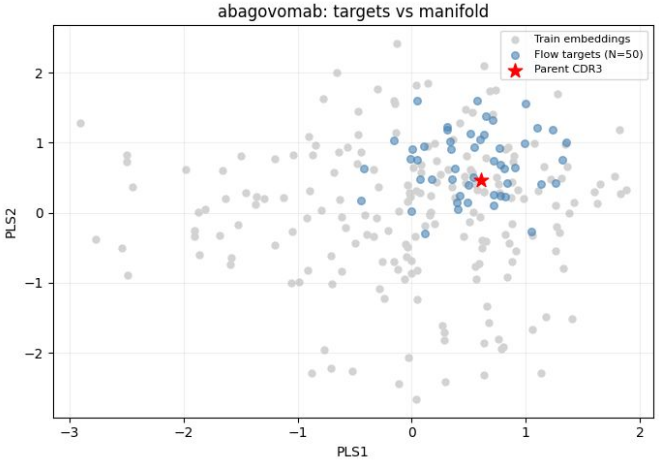
Challenges, Limitations & Solutions

1. Data scarcity

⇒ Had to use PLS reduction, frozen AbLang2, data augmentation

2. The framework conditioning seems to dominate the embedding generation

⇒ Pretrain the flow model on OAS CDR3/framework pairs and fine tune on Prophet-Ab



PLS1: HIC ρ=+0.291 SINS ρ=−0.570
PLS2: HIC ρ=−0.374 SINS ρ=−0.326
PLS3: HIC ρ=−0.388 SINS ρ=−0.100

Next Steps - Generative Antibody Modelling

- Tune generative optimization pipeline and analysis of the embedding space
  - Tune key parameters (number of candidates, temperature for AbLang2 decoding) to balance improvement and diversity
  - Check if similar substructures appear in similar embedding space

- Add additional developability scores such as chemical liabilities, $\Delta\Delta G$

- **Conditioning the generation on Antigen binding affinity and specificity**
  - Measure the binding affinity and specificity of target antigen with the optimized molecule to ensure the lead optimization doesn't reduce the binding affinity and specificity

Next Steps - Developability and Approval Prediction

- Train and evaluate sequence-based prediction model
  - Fine-tune AbLang2 on PROPHET-Ab to predict developability metrics from sequence
    - There are room for the current oracles performance to improve → Fine Tuning could help
  - Compare performance against XGBoost to assess how much developability scores can be learned without experimentally derived features

- Build and validate developability predictors
  - Evaluate correlation with experimental measurements to ensure reliable optimization signals

Singular Ask For Feedback:

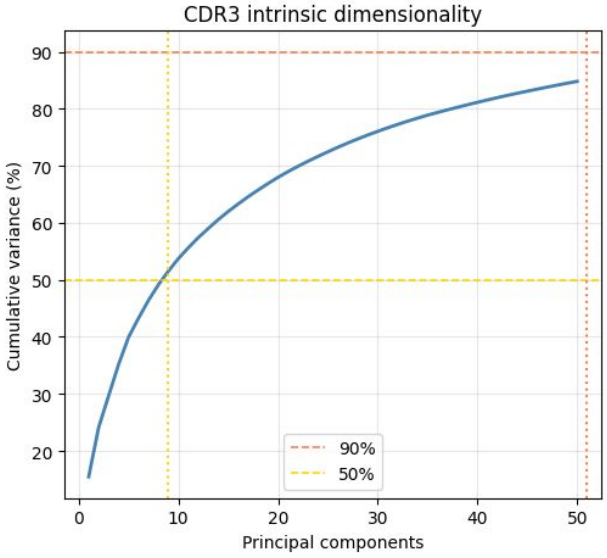
Are the biophysical properties (T_m , HIC, AC-SINS, solubility) sufficient to guide meaningful improvements in antibody developability, or should we predict additional properties to better capture real-world developability constraints?

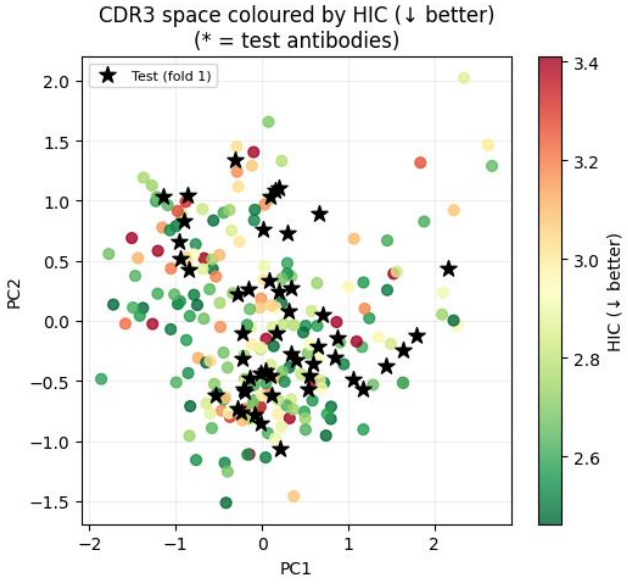
Roles

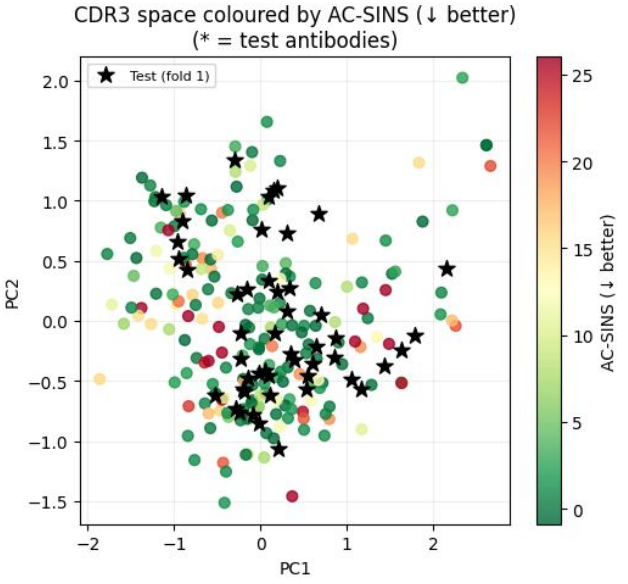
Sidhant: Built baseline models for developability metrics (ridge regressions with p-IgGen) and antibody approval (XGBoost), building fine tuned sequence based model (using AbLang2) for developability metric prediction and antibody approval

Jack: Built generative pipeline for developability optimization (dimensionality reduction of CDR3 embeddings using PLS, training a conditional flow matching model in reduced embedding space, and implementing a steered sequence generation procedure using AbLang2 guided by flow model targets)

Appendix 1


CDR3 intrinsic dimensionality
Cumulative variance (%)
Principal components
90%
50%


CDR3 space coloured by HIC (↓ better)
(* = test antibodies)
Test (fold 1)
PC1
PC2
HIC (↓ better)


CDR3 space coloured by AC-SINS (↓ better)
(* = test antibodies)
Test (fold 1)
PC1
PC2
AC-SINS (↓ better)